Feminist Economics

ISSN: 1354-5701 (Print) 1466-4372 (Online) Journal homepage: www.tandfonline.com/journals/rfec20

No Interest: The Marginalization of Women in Academic Finance

Cordelia Fine, Nitin Yadav & Carsten Murawski

To cite this article: Cordelia Fine, Nitin Yadav & Carsten Murawski (2025) No Interest: The Marginalization of Women in Academic Finance, Feminist Economics, 31:2, 263-290, DOI: 10.1080/13545701.2025.2483823

To link to this article: https://doi.org/10.1080/13545701.2025.2483823

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

View supplementary material

Published online: 11 Jun 2025.

Submit your article to this journal

Article views: 1589

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rfec20

Feminist Economics, 2025
Vol. 31, No. 2, 263–290, https://doi.org/10.1080/13545701.2025.2483823

No Interest: The Marginalization of Women in Academic Finance

Cordelia Fine, Nitin Yadav, and Carsten Murawski

ABSTRACT

Women experience significant disadvantages within the financial system, and across financial outcomes. This article investigates attention to such issues in academic finance, and its association with authorship gender. Analysis of the authorship of more than 125,000 academic finance articles (1918–2020) reveal persistent dominance of US-based men. The study uses the natural language processing technique of topic modeling to infer topics within a subset of this corpus for which abstracts were available (1988–2020). This reveals gendered patterns of publishing and an absence of attention to group-based inequalities. Hand-coding of these abstracts found that women were overrepresented as authors of the 0.78 percent of articles addressing sex/gender issues. These most commonly examined the instrumental benefits of women for others, and least commonly investigated gender barriers in finance-related domains. Results suggest that women's low representation as authors in academic finance is related to the marginalization of women's interests in knowledge production.

KEYWORDS

Gender analysis, feminist epistemology, equality, institutional bias, women and the economy, finance

JEL CODES: B54, A13, G00

HIGHLIGHTS

- Academic finance is dominated by male authors from US institutions.
- This literature marginalizes women's interests, despite men's dominance in finance.
- The most common gender-related research asks how women could benefit others.
- Women's authorship is greater on articles that include sex/gender analysis.

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

- Women's greater representation could increase knowledge that benefits women.

> My aunt . . . died by a fall from her horse when she was riding out to take the air in Bombay. The news of my legacy reached me one night about the same time that the act was passed that gave votes to women. A solicitor's letter fell into the post-box and when I opened it I found that she had left me five hundred pounds a year for ever. Of the two – the vote and the money – the money, I own, seemed infinitely the more important. (Virginia Woolf, *A Room of One's Own* 1929)

To what extent does academic finance attend to the gendered dimensions of finance? And is this attention related to gender representation among finance academics? While past research has typically been motivated by *equality* arguments for women's greater representation in academic fields (that is, focused on the importance of equal opportunities), here we contribute to growing interest in potential *epistemic* arguments (that is, focused on the importance of epistemic quality and justice in the production of knowledge).

Identifying, understanding, and reducing barriers to women's economic empowerment is considered vital for girls' and women's human rights, child welfare, and global economic growth (Scott 2020). Yet women continue to experience significant disadvantages within the financial system, and on a broad suite of financial outcomes. World Bank data finds that, globally, nearly two thirds of economies require legal reform with regards to pensions to ensure women's financial security in old age, nearly 40 percent limit women's property rights, and there are 101 economies in which women face sex-specific barriers in running a business (World Bank 2023). Women are also largely absent from the leadership of the major institutions of the financial system. For example, in January 2024, only 21.6 percent of OECD finance ministers; 13.5 percent of the heads of OECD central banks; and one of the CEOs of the world's forty largest banks by market capitalization were women.[1] Women hold only 18 percent of C-suite level leadership roles in the financial services industry globally, ranging from 11 percent in South America to 24 percent in Oceania (Deloitte 2023). In the United States, women make up just 13 percent of decision makers at venture capital firms with over $25 million in assets under management; with no women decision makers at all in 63 percent of firms (All Raise 2024).

Analyses also point to women being disadvantaged by the decisions made within global, state, and corporate financial institutions, such as the gendered effects of fiscal and monetary policies imposed as conditions of loans made by the IMF and World Bank, or the allocation of just 0.9 percent

of venture capital to European women-founded companies (Benería et al. 2000; Elson 2002; Assassi 2009; Staveren 2011; PitchBook 2024). Research has also identified barriers to women's inclusion and opportunities within male-dominated financial institutions, including on Wall Street and the City of London (McDowell 1997; Roth 2007). The UK's Office for National Statistics (Office for National Statistics 2024) reported a median gender pay gap of 34.3 percent for the UK financial services sector (excluding insurance and pensions) – the largest (of good quality estimates) across all industries.

When it comes to households, which represent another important institution within the financial system, women fare less well than men with regards to disposable income, access to bank accounts, wealth, and old-age poverty (Figure A1 in Online Appendix I), with single motherhood presenting a particularly important barrier to wealth accumulation and protection from poverty (Sierminska 2018). Substantial racial wealth gaps can intersect with gender wealth gaps, for instance giving rise to particularly poor wealth outcomes for Hispanic and Black women in the US (Chang 2012). In some countries, up to a third of women on average do not control how their own personal earned income is spent (Ortiz-Ospina, Hasell, and Roser 2018). Even in countries in which formal legal barriers to women's financial autonomy have long been dismantled, like the US and the UK, women enjoy less financial decision-making authority within households (Mader and Schneebaum 2013), and common income allocative systems within heterosexual households may entrench, rather than alleviate, lower earning women's disadvantages with regards to access to money (Kenney 2006). Women also report higher rates of economic abuse (Outlaw 2009).

Broadly speaking, finance is concerned with financial decision making and welfare within households and firms, the allocation of capital by financial institutions and markets, and the financial system overall. This brief sketch of the substantive, global gender disparities with regards to the financial system and financial outcomes therefore indicates an important role for academic finance: namely, to produce knowledge that illuminates the causes and consequences of these phenomena, and potential interventions or reforms. On the other hand, there are also reasons to anticipate that the discipline will not have paid much attention to the gendered dimensions of the financial system and financial outcomes, despite their global significance and ubiquity. Previous research has shown that finance academics, editors, and publications are very male-dominated (Hatfield and Webb 2015; Holman, Stuart-Fox, and Hauser 2018; Sherman and Tookes 2022). Indeed, at the end of the time-period on which the current research was based, we found that more than 90 percent of editors-in-chief (or equivalent) of the 171 journals considered in this article were male, and disproportionately so for higher ranked journals. We found that men held 98.5 percent of such roles for top-tier (A∗) journals and 100

percent for so-called "top-3" journals that play a particularly critical role in academic recruitment, tenure, and promotion decisions (Figure A2; Heckman and Moktan 2020).[2] As we go on to explain, there are reasons to anticipate that the male-dominance of the discipline shapes the knowledge it produces.

Influential views in feminist epistemology and philosophy of science hold that the gender of inquirers (in interaction with other dimensions of social identity) can make a difference to the production of knowledge. A central concept within such accounts is situated knowledge – the idea that a scientist's (or other inquirer's) social situation can give rise to differences in knowledge production. For example, a person's experiences and interactions are systematically shaped by their gender, contributing to group-level differences in interests, or background values and assumptions about gender relations. Feminist epistemologists have therefore made various arguments that gender representation within the social and biological sciences can matter epistemically. With regards to *what* and *how* topics are studied, this includes arguments that women scientists may be more likely to be interested in issues that pertain to the lives of girls and women, to include sex and gender variables in their research,[3] or to enjoy enhanced ability or motivation to identify or reject sexist or androcentric assumptions operating within disciplinary frameworks (Collins 1986; Longino 1990; Harding 1991; Anderson 2004; Intemann 2010). The implication of such accounts is that significant underrepresentation of women in a social science discipline is expected to lead to distortions, omissions, or limitations in scientific knowledge production, such as theories or knowledge that marginalize women's interests or activities, that present women as inferior or of interest only to the extent that they serve male interests, or that obscure gendered power relations (Anderson 2020). Economic theory, on which academic finance is founded, has been the target of many such critiques (Ferber and Nelson 1993; Barker and Kuiper 2021).

Within feminist epistemology, such arguments have often looked to specific examples and case studies for support (for example, see Crasnow [2023]). However, recent research has begun to explore associations between scientists' gender and knowledge production using quantitative analyses of large datasets. Thus, recent research has found that women are more likely than men to investigate issues of sex and gender in medicine and psychology, and topics relating to structural inequality in management science (Stockdale and Eagly 2018; Nielsen and Börjeson 2019; Sugimoto et al. 2019). Here, we apply and extend such methods to academic finance, in an initial investigation of the production of knowledge regarding gendered disparities within the financial system, including relations with authorship gender.[4]

First, we use article meta-data from a database of articles published in finance journals, to investigate women's representation as authors. We also examine whether this is moderated by time-period, in the expectation that women's authorship would increase over time. In addition, we look at associations with journal prestige and/or geographic region. We investigate the former since prior research has found persistent underrepresentation of women neuroscientists in high profile journals that are critical for career success (Shen, Shoda, and Fine 2018). Meanwhile, cross-cultural variation in women's authorship is of interest given cross-country/region variation in women's representation in other fields of study (Charles and Bradley 2009) as well as in leadership of financial institutions (Sahay and Cihak 2018). Overall, these analyses provide us with an overview of women's authorship within the discipline overall, as well as along the dimensions of time-period, prestige, time, and geographic region.

Second, we use topic modeling to map the conceptual landscape of academic finance. Briefly, topic modeling is a natural language processing technique developed for analyzing the content of large samples of text, including to extract meaningful topics from scholarly documents (Talley et al. 2011; McFarland et al. 2013; Nichols 2014; Nielsen and Börjeson 2019). In simple terms, topic modeling provides a probabilistic method for inferring the topic structure of a corpus of documents by analyzing the clustering together of relevant words. Documents – in this case individual articles – are modeled as a mix of topics, while topics are modeled as mixes of words. The optimum number of topics is identified that capture coherent themes, captured by a set of commonly co-occurring words. To the best of our knowledge, the topic structure of the academic finance literature has not been studied before. In addition, this technique enables us to explore whether gender, or inequality more broadly, is established as a topic within the discipline. We also explore whether there are gendered patterns of authorship among topics.

Finally, in order to further explore associations between women's authorship and topics, we analyze a subset of the corpus for which abstracts are available. We use hand-coding of article titles and abstracts to estimate the prevalence of sex/gender analysis within the entire corpus and to investigate whether, as has been found in some other disciplines, such research is associated with women's authorship. In addition, each of the articles was coded for the kind of sex/gender research question with which it was primarily concerned. Specifically, we classified articles as being primarily concerned with: investigating the role of gendered mechanisms (such as discriminatory laws, sex discrimination, gendered traditions, or social networks) in producing inequitable gendered financial outcomes; simply cataloguing sex differences in financial behavior or outcomes (such as in financial risk-taking or financial literacy); or investigating whether women could create value for other entities (such as increasing bank profits

by including more women customers, or improving a company's financial performance by putting more women on boards). This enabled us to go beyond mere inclusion of sex/gender analysis, and consider what kind of knowledge about sex/gender variables is produced within academic finance.

METHODS

Construction of database

To construct our database of articles, we identified the set of academic finance journals (Figure 1). Our selection is based on the Australian Business Deans Council Journal Quality List (Australian Business Deans Council 2019), a list of 2,682 academic journals curated and regularly reviewed by academic disciplinary expert panels. Each journal on the list is assigned a primary Field of Research (FOR) code, based on the Australian and New Zealand Standard Research Classification system. We selected all journals with FOR code 1502 (Banking, Finance and Investment), the code corresponding to the field of finance (which is typically considered to include the topics of banking and investment), resulting in 171 journals (Table A1). To the best of our knowledge, this is one of two lists of academic finance journals curated by a panel of academic disciplinary experts. The other list is the UK Chartered Association of Business School Academic Journal Guide which contains only sixty journals labeled as finance, all of which are included in the list we consider.

In the next step, we collected article meta-data from the Dimensions platform (Digital Science & Research Solutions Inc. 2020), including title, author, and affiliation information, for all articles published after 1900 in any of the journals on our list for which meta-data was available (Hook, Porter, and Herzog 2018). We obtained meta-data for 137,160 articles published in 139 different journals. No article meta-data was available for thirty-two journals. Articles for which meta-data could be retrieved were published between 1918 and May 2020. The cumulative number of journals in our database for which article meta-data was available for articles published in this or prior years is shown in Figure A3-A. Article count by year is shown in Figure A3-B.

We excluded 7,782 articles for which no authorship information was available. The remaining articles were associated with 263,138 authorships. Of those, 334 did not contain any information about first or last name. The average number of authorships per article is 2.03.

Assignment of authorship gender

We used a name-to-gender assignment algorithm provided by Gender API (gender-api.com 2024) to determine authorship gender in our sample. The

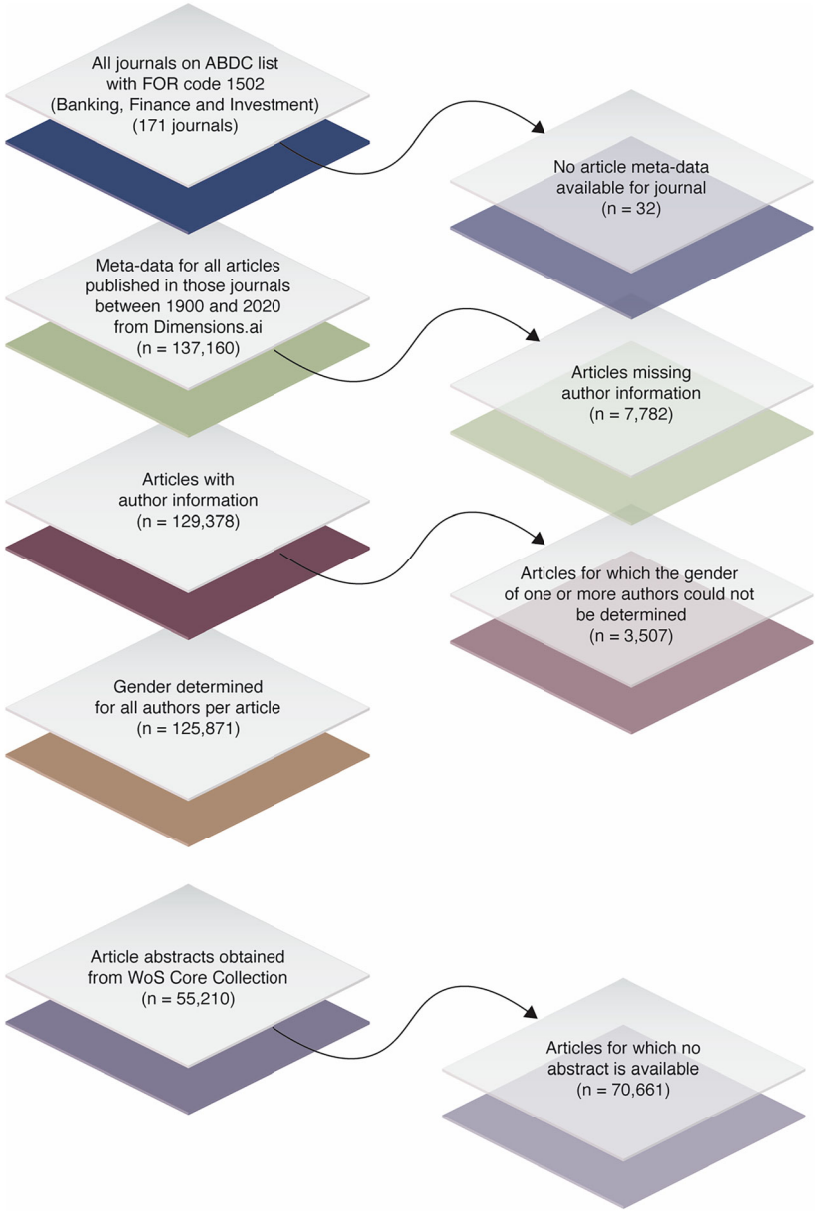

Figure 1 Construction of corpus.
Notes: We obtained article meta-data from Dimensions for all journals on a list of 171 finance journals. Meta-data were available for 137,160 articles published in 139 journals. For 129,378 of those articles, author information was available, and author gender could be determined for 125,871 articles. We were able to obtain abstracts from the Web of Science Core Collection for 55,210 articles in our database.

algorithm is based on a database of 6,084,389 validated names from 191 different countries. It predicts the gender associated with a given first name. A comparison of five name-to-gender inference services evaluated Gender API as the best-performing service (Santamaría and Mihaljević 2018).

To determine the gender of a given name, the algorithm looks up a name in the Gender API database and retrieves all occurrences of the name. If the majority of occurrences of a name in the database are labeled female, the algorithm returns the gender female, and male otherwise. The algorithm also provides an estimate of the accuracy of its gender assignment, which is computed as the percentage of the number of occurrences with a given label. For example, if a name occurs 100 times and is labeled female in seventy-five of those cases, the value of accuracy returned by the algorithm is 75.

We constructed a new variable f taking the value $a/100$ in case the predicted gender is female and $(100 - a)/100$ in case the predicted gender is male, where a is the accuracy level provided by Gender API. Thus, f takes values in the range between 0 and 1. We interpret f as the probability of an author being female (Nielsen and Börjeson 2019). We then used this variable f to construct another variable f_{weight} for each article in our database, which we computed as the average value of f of all authors of an article. This variable also ranges from 0 and 1 and can be interpreted as the estimated proportion of women among the authors of an article.

To validate the accuracy of the Gender API algorithm, we conducted a manual, web-based identification of author gender for authorships in our dataset. We proceeded as follows. There are 334 authorships in our database where no information about the first name is available, leaving 262,798 authorships with first name information (of 263,138 authorships). We then identified the unique (first name, last name) combinations in those authorships. There were 109,169 such unique combinations. We submitted those (first name, last name) combinations to Gender API. The algorithm assigned a gender to 101,476 names. Failure to assign a gender was mainly caused by missing information, in particular, availability of initial(s) only, errors in character encoding, or because a missing name was missing from the Gender API database. The names to which a gender could be assigned are associated with 249,974 of the 263,138 authorships in our database. There are 11,289 articles in our database for which no gender prediction was available for one or more authors, leaving 125,871 (of 137,160) articles for which gender information is available for all authors.

To assess the accuracy of the Gender API algorithm in inferring author gender, we constructed a validation sample of 1,092 (first name, last name) combinations (1 percent of 109,169 unique [first name, last name] combinations in our database). We determined the gender of each name in the validation sample manually, using information available on the Internet. We were able to determine the gender of 938 names. In the

remaining cases, insufficient information prevented the determination of gender, such as availability of initial(s) only or inability to determine the gender of a person from information we had access to. Gender API assigned a gender to 951 names in the validation sample.

Of the 879 names to which both Gender API and the manual web-based search assigned a gender, eleven names were classified as female by Gender API but found to be male in the manual web-search (false positive rate = 1.2 percent). Gender API classified thirty-two names as male that were found to be female in the manual search (false negative rate = 3.6 percent).

To test for statistically significant differences in gender assignment of Gender API and the web-based search, we estimated a model in which we assumed that the probability that a given name is female has a beta distribution. We then estimated separate parameters of the beta distribution for both the Gender API assignment and the web-based search, using Bayesian estimation (see below for details). The estimated difference in probabilities of a name being female, according to the two distributions, is 0.02 (95 percent highest density interval (HDI) = $[-0.02, 0.06]$). We conclude that the overall accuracy of the name-to-gender assignment algorithm is high.

To investigate the relation between gender and other variables of interest such as publication year or journal quality, we used regression analysis. All models were estimated using Bayesian techniques. Estimation is based on the Hamiltonian Monte Carlo algorithm, a very efficient Markov Chain Monte Carlo algorithm, as implemented in Stan (Carpenter et al. 2017). Given that the variable f_{weight} is constrained to the interval $[0, 1]$ and is heavily right-skewed (Figure A4 and Figure 2-A), we model this variable assuming a beta distribution with shape parameters $\alpha = \mu\nu$ and $\beta = (1-\mu)\nu$. Sampling was done with four chains and 5,000 iterations per chain.

To examine the relation between f_{weight} and publication year, we assume that $\mu = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$, where x is the (mean-centered) publication year, $\nu = e^{\gamma}$,and that the priors of μ and γ have Normal and exponential distributions, respectively.

Journal prestige

To examine gender authorship associations with journal prestige, we use journal-based metrics, in particular, the average number of citations of the annual volume for a particular journal in the year an article was published, normalized by the number of articles published in the journal in the three years prior to publication. This metric corresponds to Scopus's CiteScore metric, and we refer to it as journal impact score (JIS). We use CiteScore

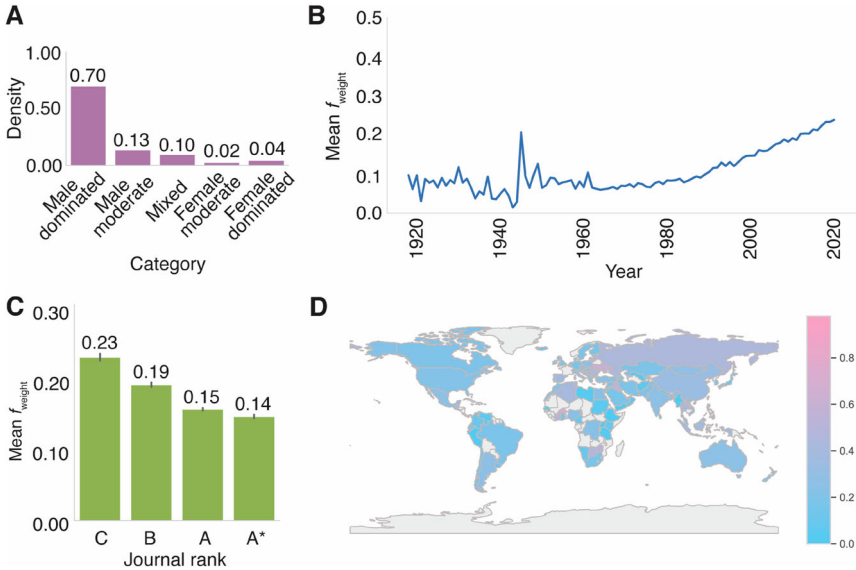

Figure 2 Female representation among authorships.
Notes: (A) Proportion of articles by category of female representation. Categories correspond to quintiles of the variable f_{weight}, which represents the estimated proportion of female authors among all authorships of an article (for example, the category "Male dominated" corresponds to values of f_{weight} between 0 and 0.2; see Methods for details). (B) Female representation over time (mean values of f_{weight} in a given year). Differences over time are statistically significant (Figure A10). (C) Female representation by journal rank categories assigned by an Australian Business Deans Council academic expert panel (see Methods for details). Differences between ranks are statistically significant (Figure A11). (D) Female representation by country. Color corresponds to the mean value of f_{weight} in a given country. Light grey indicates missing data. Differences between countries are statistically significant (Figure A17).

instead of Clarivate Analytics's Journal Impact Factor as it is available for a larger number of journals in our database. The correlation coefficient between CiteScore and Journal Impact Factor of the journals on our list for which both metrics are available is 0.93 (95 percent HDI = [0.88, 0.97]; Figure A5).

In addition to JIS, we also consider journal ranks provided by the Australian Business Deans Council as part of the Journal Quality List, an indicator of a journal's status within the discipline. In academic finance, journal ranks are commonly used in tenure and promotion processes (Heckman and Moktan 2020). Each journal in our database is assigned a rank by a panel of disciplinary experts selected by the Australian Business Deans Council. The number of journals per rank decreases monotonically in rank (Figure A6-A) while the average number of articles

per journal increases monotonically in journal rank (Figure A6-B). We attribute this pattern primarily to better availability of article meta-data for higher-quality journals. Our JIS metric as well as Journal Impact Factor, Scopus's CiteScore Percentile, and Clarivate Analytics' Journal Impact Factor Percentile metrics all increase monotonically in journal rank (Figure A7-A–D).

To examine the relation between f_{weight} and journal rank, we assume that $\mu = x$, where x is a factor variable indicating journal rank, with beta distributed priors for μ (with $\alpha = 2$ and $\beta = 2$) and gamma distributed prior for ν ($k = 2$, $\theta = 2$). To investigate the relation between f_{weight} and JIS, we assume that $\mu = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$, where x is the (mean-centered) JIS, $\nu = e^{\gamma}$,and that the priors of μ and γ have Normal (with mean equal to 0, and standard deviation equal to 2.5) and exponential ($\lambda = 1$) distributions, respectively. Sampling for this model was also done with four chains and 5,000 iterations per chain.

Geographic region

To examine geographic variation in authorship gender we determined the country of an authorship based on the author's affiliation, as indicated by the country code in the article meta-data. In case of multiple affiliations, authorship country was based on the primary affiliation listed. Country of affiliation could be determined for 114,944 of 263,138 authorships, and both gender and country of authorship could be determined for 110,025 authorships. To test for differences in gender of authorships between countries, we model f_{weight} assuming a beta distribution with shape parameters $\alpha = \mu\nu$ and $\beta = (1 - \mu)\nu$ and assume that $\mu = x$, where x is a factor variable indicating country, with beta distributed priors for μ (with $\alpha = 2$ and $\beta = 2$) and gamma distributed prior for ν ($k = 2$, $\theta = 2$). Sampling was done with two chains and 5,000 iterations per chain.

Topic modeling

In order to perform topic modeling, we used Latent Dirichlet Allocation (LDA), an unsupervised learning technique, to establish a baseline statistical image of our corpus (Blei, Ng, and Jordan 2003). LDA is a generative probabilistic model for collections of discrete data such as text corpora. Each item of a collection is modeled as a finite mixture over an underlying set of topics, and each topic is modeled as an infinite mixture over an underlying set of words. More precisely, each text reflects a Dirichlet distribution over topics, and each topic a multinomial distribution over words (Blei, Ng, and Jordan 2003; McFarland et al. 2013).

For the purposes of our study, we apply LDA to our corpus of abstracts. The Dimensions platform, our source of article meta-data, does not allow

retrieval of abstracts. We obtained the latter from the Clarivate Web of Science Core Collection (Clarivate 2022). We were able to obtain abstracts for 55,210 out of a total of 129,288 articles in our database, which are associated with 165,255 authorships.

This subset of the database, for which abstracts were available, differed from the entire database in a number of ways. First, articles for which abstracts could be obtained were published between 1988 and May 2020 (compared to 1918–2020 for the entire dataset). Second, it included articles from 104 journals (compared with 139 journals for the entire dataset). The proportion of articles published in high-quality journals is higher for articles for which we were able to obtain abstracts (A∗: 33.9 percent, A: 46.8 percent, B: 14.7 percent, C: 4.6 percent) compared to articles for which we could not obtain abstracts (A∗: 21.7 percent, A: 42.8 percent, B: 24.0 percent, C: 11.5 percent). Third, the mean of the variable f_{weight} for articles with abstracts is 0.202 (SEM = 0.001) compared to 0.145 (SEM = 0.001) for articles without abstracts. Restricting the comparison to equivalent time periods, the mean of the latter is 0.173 (SEM = 0.001; see Figure A8). In short, the subset of articles on which topic modeling (and hand coding) was based was restricted to more recent research (from 1988 onwards), and slightly higher quality, compared with the entire database.

We use an implementation of LDA provided by the Gensim package for Python (Rehurek and Sojka 2010). Before estimating the model, we preprocessed the corpus of abstracts. Preprocessing included conversion of words to lower case, removal of non-letter characters, stop words, based on the NLTK list of English stop words (Bird, Loper, and Klein 2009), tokenization, creation of bigrams and trigrams, as well as lemmatization, using the spaCy package (Honnibal and Montani 2017; Figure A9-A). The LDA model was estimated using an online variational Bayes algorithm (Hoffman, Bach, and Blei 2010) as implemented in Gensim. The (asymmetric) priors of LDA parameters α and η were learned from the data. α is a parameter of the Dirichlet distribution that influences the topic sparsity of topics in documents. Higher values suggest that documents contain a mix of most topics. η is a parameter of the multinomial distribution that is related to the probability of a word occurring in a document. The number of topics was determined by estimating the LDA model with different numbers of topics, ranging from two to fifty, running the estimation ten times for each number of topics. To compare the goodness of fit of models with different numbers of topics, we used a measure of coherence. In a topic model, the most probable words associated with a topic represent that topic. If the topic model fits well, the most probable words are semantically related. The latter can be assessed with measures of coherence. We use a measure of coherence that has been shown to correlate highly with human interpretation (measure C_V in Röder, Both, and Hinneburg [2015] as implemented in Gensim). We then

picked the model with the number of topics that has the highest coherence. Coherence was maximized at twenty-one topics (see Figure A9-B). Other model parameters were also determined by maximizing coherence. To measure the "similarity" of topics, we considered inter-topic distances, which we computed using principal component analysis to Jensen-Shannon divergences of topic distributions (Sievert and Shirley 2014).

Then, we assigned a label to each topic of the model. Each topic is given a loading indicating the topic's prevalence in the corpus, and each word is given a loading indicating its likely frequency in a topic given the prevalence of the topic. We selected the ten most prevalent words in a topic in combination with article meta-data (title, abstract, journal name) of ten randomly chosen articles for which the topic had the highest loading. One of the authors, a finance scholar (C.M.), interpreted and named each of the topics. Another one of the authors (N.Y.), also a finance scholar, corroborated the interpretation and naming. In addition, we mapped the topics to Journal of Economic Literature (JEL) codes where possible, a classification scheme for journal articles widely used in economics and adjacent disciplines. The list of topics, their prevalence in our corpus and a mapping to JEL codes is provided in Table A2. The ten words with the highest loadings within a topic as well as five randomly chosen sample articles (title, abstract, journal name) are provided in Table A3.

We complemented topic modeling with regression analysis to predict the proportion of female authors of a given article based on topic loadings. As before, we model the variable f_{weight} with a beta distribution with shape parameters $\alpha = \mu\nu$ and $\beta = (1 - \mu)\nu$ with $\mu = x$, where x is a factor variable representing the twenty-one topics. We assume Normal and gamma prior distributions for μ and ν, respectively. Sampling for this model was done with four chains and 5,000 iterations per chain. All regression models were estimated with Stan (Carpenter et al. 2017).

Coding categories and process

We identified all articles with a sex/gender word in the title and/or abstract (see Table A4 for list of sex/gender terms used in the search). Two independent coders, one with expertise in gender issues (C.F.) and the other with expertise in finance (C.M.) then coded those articles according to a procedure described in Online Appendix II (Codebook). Briefly, in the first step, the coders asserted that both a title and an abstract were present and that the article reported original research. Articles not meeting any one of these criteria were coded as Missing Information (MI). In the next step, the coders asserted that the research topic/question of the article was finance. Any articles not meeting this criterion were coded as Not Finance (NF). In the third step, it was asserted that the research reported in the article addressed a sex/gender-related question. Any article not stating

that it included a sex/gender variable was coded No Sex/Gender Analysis (NGSA).

Articles were coded according to these three inclusion criteria, followed by a set of four primary codes, developed through an iterative process involving a cursory reading of a selection of the abstracts and preliminary coding by the first author, leading to refinement of the codebook. The four possible codes were: Investigating the Role of Gendered Mechanisms in Inequitable Gendered Outcomes (Gendered Mechanisms, or GM); Catalogues of Sex Differences (CSD); The Instrumental Value of Women for Other Entities (Instrumental, or INS); and Sex/Gender as a Control Variable (SCV). Following finalization of the codebook, all articles were coded independently by the two coders. Inter-rater reliability (Cohen's kappa) was 0.636 ($p < 0.001$). Discrepancies were resolved through discussion.

RESULTS

Representation of women authors

The mean proportion of male authorships among all authorships of articles in our sample is 83.3 percent (SEM = 0.07 percent), indicating that authorships are heavily dominated by men. Indeed, 70.7 percent of articles have a proportion of male authorships of more than 80 percent while only 4.3 percent have a proportion of female authorships of more than 80 percent (Figure 2-A).

As expected, the proportion of male authorships has declined over time, from 98.6 percent in 1918, although remaining well above parity at 76.1 percent in 2020 (Figure 2-B and Figure A10). Moreover, we found that male representation increases with journal prestige. Thus, while the proportion of male authorships is 77.3 percent for the lowest ranked journals (Category C), it is 85.6 percent for the highest ranked ones (Category A∗; Figure 2-C and Figure A11). Male representation is also positively related to Scopus CiteScore, a common citation-based journal impact metric (Figure A12). The proportion of male authorships in the "top-3" journals is 89.3 percent (and 82.5 percent for the period 2010–20).

Male dominance of authorships is also moderated by geographical location: for example, female authorship approaches or exceeds parity in Albania, Armenia, Iraq, Lithuania, Moldova, and Ukraine (Figure 2-D). Nations other than the US are not well-represented within the literature, however: 38.6 percent of all authorships are located in the US (Figure A13-A) where the proportion of male authorship is 82.5 percent. Women's low representation in US institutions, and the dominance of US authorships in the literature, are therefore important contributors to women's low representation overall in the literature. The same gender and

geographical pattern was also reflected in the geographical location of the editors-in-chief of the A∗ journals who were all male, 84.6 percent residing in the US, at the endpoint of the time-period of our analysis.

Conceptual landscape of academic finance and associations with authorship gender

Next, we explored the conceptual landscape of knowledge production, and its association with authorship gender. We limit our focus to topics, in line with prior studies addressing similar research questions (Nichols 2014; Nielsen and Börjeson 2019). We conceptualize a topic as

> focus on theoretical, methodological or empirical knowledge that is shared by a number of researchers and thereby provides [. . .] a joint frame of reference for the formulation of a problem, the selection of methods or objects, the organizations of empirical data, or the interpretation of data. (Gläser, Heinz, and Havemann 2015: 1008)

Given the complex properties of topics, the relevance of using bibliometric methods, which are typically confined to a small set of paper properties such as authors, journals, references, and key terms, is limited (Gläser and Laudel 2007; Gläser, Heinz, and Havemann 2015). We used a combination of topic modeling and regression analysis to overcome this limitation.

As noted in the Methods, model fitting identified twenty-one topics as optimal. We began by considering the relation of these twenty-one topics to each other. To this end, we quantified the similarity of topic distributions using Jensen-Shannon divergence, a measure of distance between two probability distributions (Sievert and Shirley 2014). We then applied principal component analysis to inter-topic distances and located each topic along the first two principal components (Figure 3). To interpret the two principal components, we consider the differences in representative content and methods of articles belonging to topics along the first and second principal component. The main differentiating factor along the first principal component (PC1 in Figure 3) is the method of analysis. Whereas articles in topics associated with high values of the first principal component tend to be theoretical and have a higher share of mathematical content (such as finance and insurance mathematics, option pricing), articles related to topics with low values of the first principal component tend to be more empirical and less mathematical in their methods (such as corporate governance, banking, event studies). We therefore interpret the first principal component as method of analysis (theoretical/mathematical *versus* empirical). The main differentiating factor along the second principal component is the object of analysis. Whereas articles associated with topics with high values of the second principal component tend to

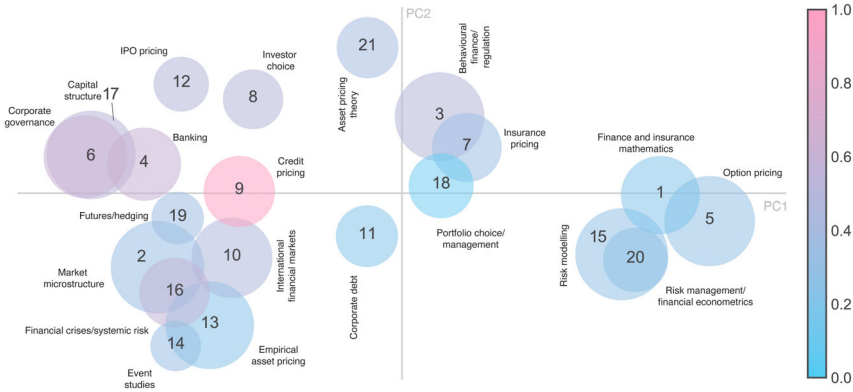

Figure 3 Topics displayed along the first two principal components of inter-topic distances.
Notes: A Latent Dirichlet Allocation model was estimated to extract semantically meaningful topics based on article abstracts ($n = 55{,}210$). Principal component analysis to Jensen-Shannon divergences of topic distributions was applied to estimate inter-topic distances. The topic coordinates in the plot are based on the first two principal components. The x-axis (PC1) represents the method of analysis (theoretical/mathematical versus empirical; high values of the first principal component tend to be theoretical and have a higher share of mathematical content). The y-axis (PC2) represents object of analysis (individuals/firms versus markets/systems; high values of the second principal component tend to focus on individuals or individual firms). Size of circles indicates prevalence of topic in the corpus. Color indicates female representation (mean of f_{weight}) within a topic.

focus on individuals or individual firms (for example, behavioral finance, investor choice, IPO pricing), articles associated with topics with low values tend to focus on markets and systems (for example, financial crises, market microstructure, international financial markets). We therefore label the second principal component as object of analysis (individuals/firms *versus* markets/systems). Notably, no topic relating to group-based inequality was identified (Table A2).

Next, we looked at associations between topics and authorship gender. Female representation varied by topic, from 30.0 percent ("credit pricing") to 14.6 percent ("portfolio choice/management"; Figure 3, Table A2, Figure A14). Female representation also varied systematically along both principal components, with lower representation in topics associated with theoretical/mathematical analyses and markets/systems, versus relatively higher representation in topics associated with empirical methods and those studying individuals/firms (Figure A15). These topics did not appear to be any less central to the discipline. (The prevalence of a topic within an entire corpus is reflected in the size of the circle in Figure 3).

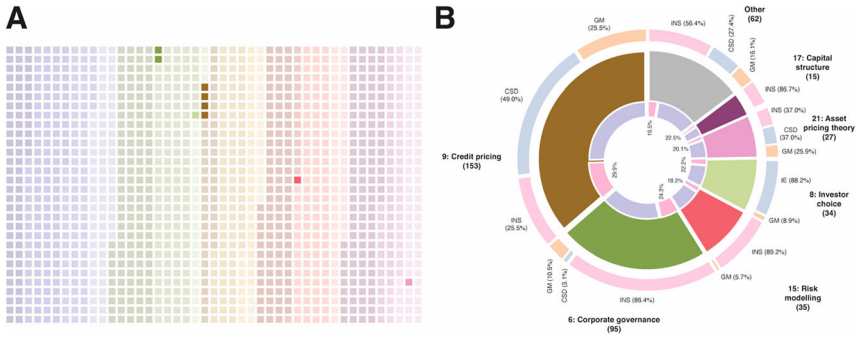

Figure 4 Results of research classification.
Notes: (A) Proportion of articles containing a sex/gender analysis in each of the topics. Each color represents a different topic. The number of squares is proportional to the prevalence of a topic in the corpus. (B) Distribution of content type by topic. The middle ring represents the prevalence of topics among all articles containing a sex/gender analysis. The innermost ring shows the proportion of male/female authorship for each of the topics. The outermost ring indicates the distribution of content types for each of the topics. Numbers in brackets below the topic names indicate the number of articles of a given topic. INS = Instrumental; CSD = Catalogues of Sex Differences; GM = Gender Mechanisms.

Sex/gender analysis in academic finance and association with authorship gender

Of the 55,210 articles with available abstracts, identification of those including a sex/gender word in the title and/or abstract yielded 627 articles (1.1 percent; see Figure 4A and 4B). Elimination of articles coded as Missing Information, Not Finance, or Sex/Gender as a Control Variable yielded 433 articles, representing 0.78 percent of the corpus used for topic modeling, and 1.06 percent in the most recent time period of 2011–20. Visual inspection indicates a relatively steep rise in sex/gender content from 2010 onwards (Figure A16).

Among these 433 articles, the most common category of sex/gender analysis research question was Instrumental (INS: $n = 222$ or 51 percent). Articles with this code investigated the instrumental consequences, value, or correlates of women (such as customers, employees, or leaders) for other entities (such as clients, business, the economy, shareholders, or risk models used for insurance purposes). This included, for example, research relating to sex-differentiated models of mortality and morbidity, or the association of board gender diversity with financial institutions' performance, efficiency, profitability, stock liquidity, or governance.

The next most common category was Catalogues of Sex Differences (CSD: $n = 139$ or 32 percent). Articles with this code investigated male/female differences in behavior (for example, risk preferences,

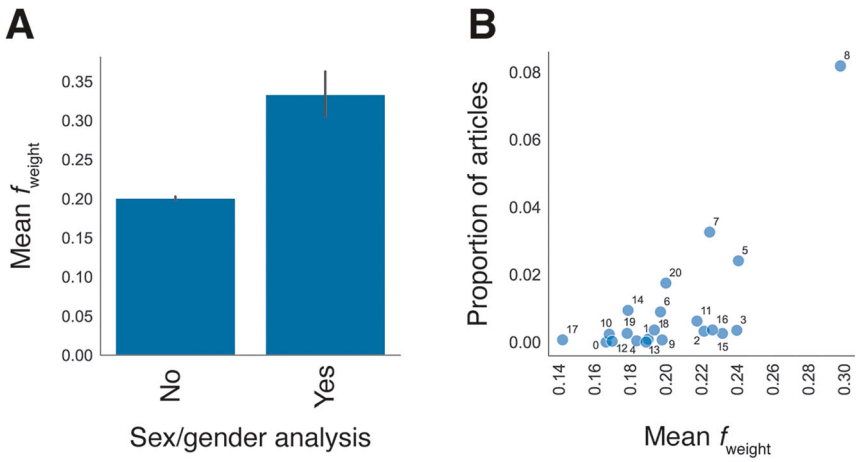

Figure 5 Female representation (mean of f_{weight}) and sex/gender content.
Notes: (A) Mean female representation for articles with and without a sex/gender analysis. Vertical bars indicate standard error of the mean. (B) Mean female representation in a topic and proportion of articles with sex/gender analysis in the topic, by topic.

financial knowledge, trading behavior) and/or finance-related outcomes (for example, portfolio returns). The least common category of sex/gender analysis research question was Gender Mechanisms (GM: $n = 72$, 17 percent). Articles with this code investigated formal or informal gendered mechanisms and/or their contribution to finance-related outcomes. Examples include investigations of sex discrimination in lending decisions, or the association of female leadership with employee gender wage gaps.

To examine the link between the gender of authors of articles and their content, we analyzed the relation between the proportion of female authors of an article and whether an article contained sex/gender analysis. The mean proportion of female authorship of articles containing a sex/gender analysis (mean of f_{weight}) is 0.33 (SEM = 0.02) and 0.20 (SEM = 0.001) for those without (Figure 5-A). To test whether this difference is significant, we regressed f_{weight} of articles on a categorical variable indicating whether an article contained sex/gender analysis. The difference in means of the posterior distributions of the categories is 0.13 (95 percent CI = [0.11, 0.16]). This means that the proportion of female authors for articles containing a sex/gender analysis was 12 percentage points (or 66 percent) higher than that of articles without such analysis.

Finally, we examined the relation between female authorships of articles and sex/gender content within topics. To this end, we correlated the proportion of female authorship with the proportion of articles containing a sex/gender analysis within topics. We found that topics with a higher

proportion of female authorship (higher mean of f_{weight}) were associated with a higher proportion of articles containing a sex/gender analysis (correlation $\rho = 0.45$, 95 percent HDI = [0.02, 0.78], $P(\rho \geq 0) = 0.97$). This means that topics with a higher representation of female authors were more likely to contain articles investigating sex/gender-related issues (Figure 5-B).

DISCUSSION

In every country in the world, women fare poorly relative to men in the institutions that make up the financial system, and on every measure of financial outcomes. To what extent does academic finance pursue knowledge that could illuminate and address these issues, and are there reasons to think that such pursuit may be related to the gender representation of authors within the discipline? To investigate these questions, this study analyzed the authorship gender of more than 129,288 academic finance articles published between 1918 and 2020, disaggregating by geographic region, time period, and journal prestige. We then used natural language processing to identify the conceptual landscape of the discipline, and association with authorship gender, using a subset of 55,210 articles for which abstracts were available. Using the same subset of articles, we used hand coding to explore the frequency of articles incorporating sex/gender analysis, disaggregated by whether they examined instrumental benefits of women for other entities, catalogued sex differences on one or more variable, or investigated gendered mechanisms giving rise to women's disadvantage in finance-related domains.

Our first main set of findings build on prior research showing male dominance within the field (Hatfield and Webb 2015; Holman, Stuart-Fox, and Hauser 2018). Our analysis confirms the male dominance in authorship and chief editorships, while additionally contextualizing this male dominance as being intensified within high prestige journals, and strongest and most impactful in US institutions. Although our findings are concerned with the epistemic consequences of women's representation within academic finance, rather than its causes, we note that prior research on other disciplines indicates gender bias within prestige journals, both in relation to content and women's representation (Nielsen 2017; Shen, Shoda, and Fine 2018). This points to a useful line of inquiry for academic finance, in light of the current findings.

In addition, we found that male dominance of authorship was contingent on geographic region, with gender parity or even women's dominance observed in a small number of countries, mostly located in Eastern Europe. This points to another potential line of investigation with regards to understanding why academic finance has remained so strongly male dominated in countries such as the US. Some have argued that persistent

gender imbalances in science, technology, engineering, and mathematics fields observed in advanced economies are not a matter of concern, because they reflect women and men being better able to pursue their real occupational preferences in contexts of greater economic security (Stern and Madison 2022). However, historical and sociological research points to far more complex explanations of why some fields are strongly male-dominated in rich, Western countries while being more gender-balanced elsewhere (Charles and Bradley 2009). For example, research has pointed to the observable social (re)construction of computer science as masculine as opportunities for pay and prestige increased (Ensmenger 2015; Abbate 2017), girls' and minority boys' lack of identification with tech identities (Budge et al. 2023), stronger gender-typing of math as masculine in post-industrial economies (Breda et al. 2020), and effects of post-industrial economic structure and cultural values on gender norms and their expression (Charles and Grusky 2004; Block et al. 2022). Future work could usefully investigate the role of such factors in the persistent male dominance of academic finance in the US, and other countries in which the rich opportunities wrought by financialization may offer part of the explanation for particularly strong gender-typing as a masculine domain.

Our second main set of findings concerns the conceptual landscape of academic finance. Topic modeling identified twenty-one topics within the discipline, demonstrating knowledge production across a wide range of lines of inquiry, such as corporate governance, financial crises/systemic risk, finance and insurance mathematics, and option pricing. However, no topic relating to group-based inequalities or disadvantages within the financial system was identified (Table A2). This is in contrast with topic modeling of the cognate field of management science (Nielsen and Börjeson 2019). These authors identified group-based inequalities as a topic (termed Structural Inequality), encompassing study of "Prevailing inequalities in organizations related to gender, race, ethnicity and age" and for which "women" and "gender" were the most common words (see Online Appendix I). The absence of a similar topic within academic finance – namely, prevailing inequalities in the financial system related to gender and other attributes – is notable. Values inevitably play a significant role in scientific research, including with regards to decisions as to what topics are studied, what the aims of research are, and what facets of a phenomenon are considered important to model (Douglas 2009; Elliott 2017). There are no value-neutral ways to make these decisions. Our findings suggest that, overall, values of equality and justice are not prioritized in research questions selected within academic finance.

We also examined the association of topics with authorship gender, finding that women's authorship was somewhat higher among topics concerned with individuals and firms versus markets, and those that deployed empirical versus mathematical/theoretical methods. We can only

speculate as to the reasons for these gendered patterns, which do not appear to be explained in terms of women being more likely to publish in more marginal topics within the discipline. Possibly, these findings may be part of a general pattern whereby fields assumed to require innate brilliance have fewer women and racial minorities (Leslie et al. 2015), or women tend to work on more people-focused topics (Nielsen and Börjeson 2019).

Our third set of findings further explored the content of academic finance and associations with authorship gender. In keeping with the absence of a topic relating to structural inequality within academic finance, we found that only a small percentage of articles incorporated sex/gender analysis, approximately 1 in 100, even in the most recent decade. Importantly, the subset of the database on which these findings are based (comprising articles for which we were able to extract abstracts for coding) were published between 1988 and 2020, and were skewed towards higher prestige journals. This percentage is therefore likely an overestimation of the representation of sex/gender analysis in the corpus as a whole, instead offering an estimate of the extent to which such research appears in more recent, better-quality research within the discipline. However, there were many federal legislative changes during the second wave feminist movement in the US, such as the Equal Pay Act (1963), the Federal Civil Rights Act (1964), the Equal Credit Opportunity Act (1974), and the Pregnancy Discrimination Act (1978), and it is possible that these legislative changes inspired corresponding research on their effects prior to 1988.

Further coding of article abstracts revealed that, among these articles, the most common sex/gender analysis (51 percent) involved assessing the instrumental value of women for other entities (such as banks, firm boards, or insurance companies). Such research is of social value. For instance, we all have an interest in well-governed companies, efficient financial institutions, and accurate modeling for public policy. In addition, data indicating instrumental benefits to organizations of the inclusion of women in top management teams can ameliorate concerns about women's business leadership abilities (Kulik and Metz 2015). However, such research can implicitly make women's (and other underrepresented groups') inclusion contingent on demonstration of instrumental benefits (Sinclair 2000). More generally, it cannot be said to be directly concerned with women's interests or gender-related inequalities and disadvantages.

We found that the next most common type of research question (32 percent) involved cataloguing differences between men and women. Such findings can yield important insights, including documenting gender gaps in financial outcomes, or testing gender stereotypes about women's financial behavior against data (Nelson 2017; Fine 2018). However, such research can also perpetuate gender stereotypes through various

forms of confirmation bias (Nelson 2014). This research approach also cannot produce evidence of social and structural contributions to gender inequalities. Accordingly, it cannot challenge assumptions that inequalities of finance-related outcomes are attributable to inherent sex differences in preferences or skills.

The least common kind of research question (17 percent) concerned gender mechanisms associated with inequitable financial outcomes. Thus, our findings indicate that between 1988 and 2020, 0.19 percent of articles investigated such issues. While of course there is no "right" percentage of this kind of sex/gender analysis research against which to compare this result, it is worth noting that there was considerable international political attention on gender inequality within the financial system during this time-period, marked by the publication of the Beijing Declaration and Platform for Action by UN Women in 1995, adopted by 189 countries. This report highlighted the importance of governments, central banks, national development banks, and private banking institutions for increasing women's participation in financial decision making and access to financial services and markets, including via supervisory and regulatory reforms within financial institutions (United Nations 1996). Moreover, it is a reflection of the values operating within academic finance that the production of knowledge to establish the instrumental value of women (for other entities) was three times more common than research into gender mechanisms producing and maintaining women's financial disadvantages. Or, to offer a further comparison, between 1988 and 2020 there were 5,036 articles in our dataset concerned with the topic of option pricing – a specific type of financial security used primarily for risk management and financial speculation, mainly by finance professionals – compared with 72 articles (equal to 1.4 percent of 5,036) investigating finance-related gender barriers faced by a group representing approximately half the global population.

We additionally found that, overall and across most topics, women's authorship was greater on articles including sex/gender analysis. This supports philosophical arguments that gender representation is relevant for knowledge production, and extends existing empirical findings, linking authorship gender and sex/gender analysis, to the discipline of academic finance. Our findings provide support to claims that gender diversity can broaden the perspectives brought to bear on a discipline, and give rise to knowledge production that more equitably benefits the wider community (Harding 1991; Nielsen and Börjeson 2019). However, it is important to point out that feminist philosophical perspectives on science typically propose deeper epistemic benefits to the inclusion of women (and other marginalized groups) than simply a larger head count of researchers pursuing topics relevant to disadvantaged populations. For example, qualitative case studies suggest causal links between an influx of women into a discipline, and challenges to specific assumptions and

frameworks that influence an entire field of inquiry (Haraway 2006). In this regard, finance appears to fall well behind the cognate discipline of economics, this journal being just one manifestation of now well-developed feminist perspectives (Ferber and Nelson 1993; Rolin 2012).

In conclusion, women are poorly represented within academic finance: both as authors and chief editors, particularly among the highest prestige journals. Furthermore, women's interests and experiences within the financial system, and their financial outcomes, are poorly represented in the aims and approaches in the production of knowledge in the discipline. Our data further suggest that the two findings are related, and that there would likely be beneficial epistemic effects of greater inclusion of women finance academics, and greater geographic diversity (He, Fu, and Lucey 2021), for demarginalizing research questions relating to sex and gender, and perhaps also issues of group-based inequality more generally.

Cordelia Fine
The University of Melbourne - History and Philosophy of Science, Melbourne, Victoria, Australia
email: cfine@unimelb.edu.au

Nitin Yadav
The University of Melbourne - Finance, Melbourne, Victoria, Australia
email: nitin.yadav@unimelb.edu.au

Carsten Murawski
The University of Melbourne - Finance, Melbourne, Victoria, Australia,
email: carstenm@unimelb.edu.au

NOTES ON CONTRIBUTORS

Cordelia Fine is Professor in the History & Philosophy of Science programme in the School of Historical & Philosophical Studies at the University of Melbourne. Her work analyses scientific and popular biological explanations of behavioral sex differences and workplace gender inequalities, explores the effects of gender-related attitudes and biases on judgements and decision making, and contributes to debates about workplace gender equality. She is the author of *Delusions of Gender: How our Minds, Society, and Neurosexism Create Difference* and *Testosterone Rex: Myths of Sex, Science & Society*, which won the Royal Society Insight Investment Science Book Prize.

Nitin Yadav is a senior lecturer in computational finance in the Department of Finance at the University of Melbourne. His work involves the study of the complexity of decision making (in particular NP-Hard problems) and the application of techniques from computer science and artificial

intelligence to problems in finance. Nitin's other research interests lie in the areas of service composition, intelligent agent systems, and use of formal techniques for testing agent designs. Nitin obtained his PhD in Computer Science from RMIT University in 2014. He also holds a Bachelor's degree in Medicine and Surgery, a Master's degree in Computer Science with a specialization in artificial intelligence, and a Master's degree in Finance.

Carsten Murawski is Director of the Centre for Brain, Mind and Markets in the Department of Finance at the University of Melbourne. The current focus of his work is on determining in what ways information processing constraints in the brain affect decision making, how they lead to phenomena such as cognitive biases, and in what ways decision making can be improved. He is particularly interested in complex problem solving, learning about uncertainty, social interaction and meta-decision making.

ACKNOWLEDGMENTS

The authors are grateful to Justin Shearer for facilitating access to Dimensions and Clarivate databases. They are also grateful to Julie Nelson and Karin Verspoor for their helpful comments on earlier versions of the manuscript, and to Isaac Miles-Fine and Layla Azura Amir Zain for research assistance.

NOTES

[1] Gender of leaders was determined manually for each of the respective institutions using web-based search.

[2] These three journals are the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics*. At time of submission, we note that the editors-in-chief of the first and third of these journals were women.

[3] By "sex variables" we refer to disaggregation of data on the basis of sex (male or female), or investigation of sex-related variables, such as sex-differentiated genes and hormones or sex-specific experiences (such as pregnancy). By "gender variables" we refer to investigation of variables related to cultural norms, practices, meanings, and so on, associated with sex in a particular culture, such as gender stereotypes or discriminatory laws.

[4] Although our article uses the terms *female* and *male* synonymously with *woman* and *man*, any transgender authors are categorized in accordance with the gender inferred from their name by the Gender-API.

SUPPLEMENTAL DATA

Supplemental data for this article can be accessed online at https://doi.org/10.1080/13545701.2025.2483823.

REFERENCES

Abbate, Janet. 2017. *Recoding Gender: Women's Changing Participation in Computing*. Cambridge, MA: MIT Press.

All Raise. 2024. "About." *All Raise*. https://www.allraise.org/about.

Anderson, Elizabeth. 2004. "Uses of Value Judgments in Science: A General Argument, with Lessons from a Case Study of Feminist Research on Divorce." *Hypatia* 19(1): 1–24.

Anderson, Elizabeth. 2020. "Feminist Epistemology and Philosophy of Science." In *The Stanford Encyclopedia of Philosophy*, edited by Edward N. Zalta. https://plato.stanford.edu/archives/spr2020/entries/feminism-epistemology/

Assassi, Libby. 2009. *The Gendering of Global Finance*. Basingstoke: Palgrave Macmillan.

Australian Business Deans Council. 2019. *The 2019 ABDC Journal Quality List*. Australian Business Deans Council. https://abdc.edu.au/research/abdc-journal-list/2019-review/.

Barker, Drucilla K. and Edith Kuiper. 2021. "Feminist Economics." In *The Routledge Handbook of Feminist Philosophy of Science*, edited by Sharon L. Crasnow, and Kristen Intemann, 355–67. New York: Routledge.

Benería, Lourdes, Maria Floro, Caren Grown, and Martha MacDonald. 2000. "Introduction: Globalization and Gender." *Feminist Economics* 6(3): vii–xviii.

Bird, Steven, Edward Loper, and Ewan Klein. 2009. *Natural Language Processing with Python*. Beijing: O'Reilly Media.

Blei, David M., Andrew Y. Ng, and Michael I. Jordan. 2003. "Latent Dirichlet Allocation." *Journal of Machine Learning Research* 3: 993–1022.

Block, Katharina, Maria I. T. Olsson, Toni Schmader, Colette Van Laar, Sarah E. Martiny, Carolin Schuster, Sanne Van Grootel, Loes Meeussen, and Alyssa Croft. 2022. "The Gender Gap in the Care Economy is Larger in Highly Developed Countries: Socio-Cultural Explanations for Paradoxical Findings." Preprint. PsyArXiv. doi:10.31234/osf.io/k6g5d.

Breda, Thomas, Elyès Jouini, Clotilde Napp, and Georgia Thebault. 2020. "Gender Stereotypes Can Explain the Gender-Equality Paradox." *Proceedings of the National Academy of Sciences* 117(49): 31063–69.

Budge, Jason, Maria Charles, Yariv Feniger, and Halleli Pinson. 2023. "The Gendering of Tech Selves: Aspirations for Computing Jobs Among Jewish and Arab/Palestinian Adolescents in Israel." *Technology in Society* 73: 102245.

Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. "Stan: A Probabilistic Programming Language." *Journal of Statistical Software* 76(1): 1–32.

Chang, Mariko Lin. 2012. *Shortchanged: Why Women Have Less Wealth and What Can Be Done about It*. New York: Oxford University Press.

Charles, Maria and Karen Bradley. 2009. "Indulging Our Gendered Selves? Sex Segregation by Field of Study in 44 Countries." *American Journal of Sociology* 114(4): 924–76.

Charles, Maria and David B. Grusky. 2004. *Occupational Ghettos: The Worldwide Segregation of Women and Men*. Stanford, CA: Stanford University Press.

Clarivate. 2022. "Web of Science Core Collection." Clarivate. https://clarivate.com/webofsciencegroup/solutions/web-of-science-core-collection/.

Collins, Patricia Hill. 1986. "Learning from the Outsider Within: The Sociological Significance of Black Feminist Thought." *Social Problems* 33(6): S14–32.

Crasnow, Sharon. 2023. "Feminist Perspectives on Science." In *The Stanford Encyclopedia of Philosophy*, edited by Edward N. Zalta, and Uri Nodelman. https://plato.stanford.edu/archives/fall2023/entries/feminist-science/

Deloitte. 2023. "Women in Financial Services Leadership, 2023 Global Update." *Deloitte Insights*. https://www2.deloitte.com/uk/en/insights/industry/financial-services/women-leaders-financial-services.html.

Digital Science & Research Solutions Inc. 2020. *Dimensions*. Digital Science & Research Solutions Inc. https://app.dimensions.ai/discover/publication.

Douglas, Heather E. 2009. *Science, Policy, and the Value-Free Ideal*. Pittsburgh, PA: University of Pittsburgh Press.

Elliott, Kevin Christopher. 2017. *A Tapestry of Values: An Introduction to Values in Science*. New York: Oxford University Press.

Elson, Diane. 2002. "Gender Justice, Human Rights, and Neo-Liberal Economic Policies." In *Gender Justice, Development, and Rights*, edited by Maxine Molyneux, and Shahra Razavi, 78–114. Oxford: Oxford University Press.

Ensmenger, Nathan. 2015. "'Beards, Sandals, and Other Signs of Rugged Individualism': Masculine Culture within the Computing Professions." *Osiris* 30(1): 38–65.

Ferber, Marianne A. and Julie A. Nelson, eds. 1993. *Beyond Economic Man: Feminist Theory and Economics*. Chicago: University of Chicago Press.

Fine, Cordelia. 2018. *Testosterone Rex: Myths of Sex, Science, and Society*. New York: Norton.

gender-api.com. 2024. "Gender API." Gender-API. https://gender-api.com/.

Gläser, Jochen, Michael Heinz, and Frank Havemann. 2015. *Epistemic Diversity as Distribution of Paper Dissimilarities*. Dordrecht, Netherlands: International Society for Scientometrics and Informetrics.

Gläser, Jochen and Grit Laudel. 2007. "The Social Construction of Bibliometric Evaluations." In *The Changing Governance of the Sciences*, edited by Richard Whitley and Jochen Gläser, 101–23. Dordrecht: Springer Netherlands.

Haraway, Donna Jeanne. 2006. *Primate Visions: Gender, Race, and Nature in the World of Modern Science*. New York: Routledge.

Harding, Sandra G. 1991. *Whose Science? Whose Knowledge? Thinking from Women's Lives*. Ithaca, NY: Cornell University Press.

Hatfield, Patricia and Shelly Webb. 2015. "The Role of Gender in Academic Finance Journals: An Exploratory Study." *American Journal of Business Education* 8(4): 249–58.

He, Feng, Tong Fu, and Brian Lucey. 2021. "Geographic Diversity in Academic Finance Editorial Boards – A Discussion." *Finance Research Letters* 43: 102006.

Heckman, James J. and Sidharth Moktan. 2020. "Publishing and Promotion in Economics: The Tyranny of the Top Five." *Journal of Economic Literature* 58(2): 419–70.

Hoffman, Matthew, Francis R. Bach, and David M. Blei. 2010. "Online Learning for Latent Dirichlet Allocation." In *Advances in Neural Information Processing Systems 23*, edited by J. D. Lafferty, C. K. I. Williams, J. Shawe-Taylor, R. S. Zemel, and A. Culotta, 1–9. San Diego: NeurIPS Proceedings.

Holman, Luke, Devi Stuart-Fox, and Cindy E. Hauser. 2018. "The Gender Gap in Science: How Long Until Women Are Equally Represented?" *PLOS Biology* 16(4): e2004956.

Honnibal, Matthew and Ines Montani. 2017. "spaCy 2: Natural Language Understanding with Bloom Embeddings, Convolutional Neural Networks and Incremental Parsing."

Hook, Daniel W., Simon J. Porter, and Christian Herzog. 2018. "Dimensions: Building Context for Search and Evaluation." *Frontiers in Research Metrics and Analytics* 3: 23.

Intemann, Kristen. 2010. "25 Years of Feminist Empiricism and Standpoint Theory: Where Are We Now?" *Hypatia* 25(4): 778–96.

Kenney, Catherine T. 2006. "The Power of the Purse: Allocative Systems and Inequality in Couple Households." *Gender & Society* 20(3): 354–81.

Kulik, Carol T. and Isabel Metz. 2015. "Women at the Top: Will More Women in Senior Roles Impact Organizational Outcomes?" In *The Oxford Handbook of Strategy Implementation, Vol. 1*, edited by Michael A. Hitt, Susan E. Jackson, Salvador Carmona,

Leonard Bierman, Christina E. Shalley, and Douglas Michael Wright, 239–82. Oxford: Oxford University Press.

Leslie, Sarah-Jane, Andrei Cimpian, Meredith Meyer, and Edward Freeland. 2015. "Expectations of Brilliance Underlie Gender Distributions Across Academic Disciplines." *Science* 347(6219): 262–65.

Longino, Helen E. 1990. *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton, NJ: Princeton University Press.

Mader, Katharina and Alyssa Schneebaum. 2013. "The Gendered Nature of Intra-Household Decision Making in and across Europe." WU Vienna University of Economics and Business Working Paper Series 157. WU Vienna University of Economics and Business, Vienna.

McDowell, Linda. 1997. *Capital Culture: Gender at Work in the City*. Oxford: Blackwell.

McFarland, Daniel A., Daniel Ramage, Jason Chuang, Jeffrey Heer, Christopher D. Manning, and Daniel Jurafsky. 2013. "Differentiating Language Usage Through Topic Models." *Poetics* 41(6): 607–25.

Nelson, Julie A. 2014. "The Power of Stereotyping and Confirmation Bias to Overwhelm Accurate Assessment: The Case of Economics, Gender, and Risk Aversion." *Journal of Economic Methodology* 21(3): 211–31.

———. 2017. *Gender and Risk-Taking: Economics, Evidence and Why the Answer Matters*. Abingdon: Routledge.

Nichols, Leah G. 2014. "A Topic Model Approach to Measuring Interdisciplinarity at the National Science Foundation." *Scientometrics* 100(3): 741–54.

Nielsen, Matthias Wullum. 2017. "Scientific Performance Assessments Through a Gender Lens." *Science & Technology Studies* 31: 2–30.

Nielsen, Mathias Wullum and Love Börjeson. 2019. "Gender Diversity in the Management Field: Does It Matter for Research Outcomes?" *Research Policy* 48(7): 1617–32.

Office for National Statistics. 2024. "Gender Pay Gap." Office for National Statistics. https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworkinghours/datasets/annualsurveyofhoursandearningsashegenderpaygaptables.

Ortiz-Ospina, Esteban, Joe Hasell, and Max Roser. 2018. "Economic Inequality by Gender." OurWorldInData. https://ourworldindata.org/economic-inequality-by-gender.

Outlaw, Maureen. 2009. "No One Type of Intimate Partner Abuse: Exploring Physical and Non-Physical Abuse Among Intimate Partners." *Journal of Family Violence* 24(4): 263–72.

PitchBook. 2024. "European VC Female Founders Dashboard." PitchBook. https://pitchbook.com/news/articles/the-european-vc-female-founders-dashboard.

Rehurek, Radim and Petr Sojka. 2010. "Software Framework for Topic Modelling with Large Corpora." In *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, 45–50. Valletta, Malta: ELRA. http://is.muni.cz/publication/884893/en.

Röder, Michael, Andreas Both, and Alexander Hinneburg. 2015. "Exploring the Space of Topic Coherence Measures." In *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining*, 399–408. New York: Association for Computing Machinery. https://doi.org/10.1145/2684822.2685324.

Rolin, Kristina. 2012. "Feminist Philosophy of Economics." In *Philosophy of Economics*, edited by Uskali Mäki, 199–217. Amsterdam: Elsevier.

Roth, Louise Marie. 2007. "Women on Wall Street: Despite Diversity Measures, Wall Street Remains Vulnerable to Sex Discrimination Charges." *Academy of Management Perspectives* 21(1): 24–35.

Sahay, Ratna and Martin Cihak. 2018. "Women in Finance: A Case for Closing Gaps." *Staff Discussion Notes* 18(5): 1. doi:10.5089/9781484375907.006.

Santamaría, Lucía and Helena Mihaljević. 2018. "Comparison and Benchmark of Name-to-Gender Inference Services." *PeerJ Computer Science* 3: e156.

Scott, Linda M. 2020. *The Double X Economy: The Epic Potential of Women's Empowerment*. New York: Farrar, Straus and Giroux.

Shen, Yiqin Alicia, Yuichi Shoda, and Ione Fine. 2018. "Too Few Women Authors on Research Papers in Leading Journals." *Nature* 555(7695): 165.

Sherman, Mila Getmansky and Heather E. Tookes. 2022. "Female Representation in the Academic Finance Profession." *Journal of Finance* 77(1): 317–65.

Sierminska, Eva. 2018. "The 'Wealth-Being' of Single Parents." In *The Triple Bind of Single-Parent Families: Resources, Employment and Policies to Improve Wellbeing*, edited by Laurie C. Maldonado, and Rense Nieuwenhuis, 51–80. Bristol: Policy Press.

Sievert, Carson and Kenneth Shirley. 2014. "LDAvis: A Method for Visualizing and Interpreting Topics." In *Proceedings of the Workshop on Interactive Language Learning, Visualization, and Interfaces*, 63–70. Baltimore, Maryland, USA: Association for Computational Linguistics.

Sinclair, Amanda. 2000. "Critical Diversity Management Practice in Australia: Romanced or Co-Opted?" In *Handbook of Workplace Diversity*, edited by Alison Konrad, Pushkala Prasad, and Judith Pringle, 511–30. London: Sage.

Staveren, Irene. 2011. "From Gender as Exogenous to Gender as Endogenous in the New Economics." In *New Economics as Mainstream Economics*, edited by P. Arestis, and M. Sawyer, 116–53. London: Palgrave Macmillan.

Stern, Charlotta and Guy Madison. 2022. "Sex Differences and Occupational Choice Theorizing for Policy Informed by Behavioral Science." *Journal of Economic Behavior & Organization* 202(October): 694–702.

Stockdale, Margaret S. and Alice H. Eagly. 2018. "Beyond Representation of Women in I-O to Producing Gender-Inclusive Knowledge." *Industrial and Organizational Psychology* 11(3): 448–55.

Sugimoto, Cassidy R, Yong-Yeol Ahn, Elise Smith, Benoit Macaluso, and Vincent Larivière. 2019. "Factors Affecting Sex-Related Reporting in Medical Research: A Cross-Disciplinary Bibliometric Analysis." *Lancet* 393(10171): 550–9.

Talley, Edmund M, David Newman, David Mimno, Bruce W Herr, Hanna M Wallach, Gully A P C Burns, A. G. Miriam Leenders, and Andrew McCallum. 2011. "Database of NIH Grants Using Machine-Learned Categories and Graphical Clustering." *Nature Methods* 8(6): 443–44. doi:10.1038/nmeth.1619.

United Nations. 1996. *Report of the Fourth World Conference on Women*. New York: United Nations. https://www.un.org/womenwatch/daw/beijing/pdf/Beijing%20full%20report%20E.pdf.

World Bank. 2023. *Women, Business and the Law 2023*. Washington, DC: World Bank. https://openknowledge.worldbank.org/bitstream/handle/10986/35094/9781464816529.pdf.